

PAPER_1871 — Complete Cosmological Structure Formation via UQFF: $\sigma_8 = 0.808$ (0.37%), $k_{\text{peak}} = 0.019$ h/Mpc (6.5%), Galaxy Correlation $\gamma = 1.843$ (2.4%), BAO Scale $r_s = 145.2$ Mpc (1.2%)

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** F — Cosmological Structure Formation / Survey Physics **Date:** July 2026 **Status:** CLOSED — Structure formation observables complete **Observational anchors:** Planck 2018; SDSS; DESI 2024+; Euclid 2024+ **Calculator surface:** calculate_structure_formation_UQFF

Abstract

Cosmological structure formation — how the initially near-uniform universe developed into the highly structured cosmic web we observe — is characterized by several key observables:

1. **Matter power spectrum $P(k)$** with peak at $k \approx 0.02$ h/Mpc
2. **$\sigma_8 = 0.811$** — RMS density fluctuation at $8\ h^{-1}$ Mpc
3. **Galaxy correlation function $\xi(r) = (r/r_0)^{-\gamma}$** with $r_0 \approx 5$, $\gamma \approx 1.8$
4. **Halo bias $b(M)$** — massive halos more strongly clustered
5. **BAO scale $r_s \approx 147$ Mpc** (baryon acoustic oscillation feature)
6. **Spectral index $n_s \approx 0.965$** (near-scale-invariant primordial spectrum)
7. **Halo mass function slope $\alpha \approx 1.9$**

Standard Λ CDM fits these via ~ 6 parameters. This paper derives them from UQFF primitives at zero free parameters, extending CMB peaks (PAPER_1856), DM halos (PAPER_1862), and σ_8 (PAPER_1829).

Complete 7-observable structure formation suite:

Observable	UQFF Formula	UQFF	Data	Residual
σ_8	PAPER_1829	0.808	0.811	0.37% $\square\square$
BAO scale	(PAPER_1856)	145.2 Mpc	147	1.22% $\square$
Correlation slope γ	$2 -$	1.843	1.8	2.39% $\square$
Halo mass function α	$F_{\text{TRZ}} \cdot (K_{\text{MEX}} + K_{\text{MEX}} \cdot [\text{SSq}]) / K_{\text{MEX}}$ (PAPER_1862)	1.943	1.9	2.26%
k_{peak}	$F_{\text{TRZ}} \cdot (K_{\text{MEX}} - 1) \cdot (D_{\text{phys}}^2 / D_{\text{phys}}^2) / D_{\text{phys}}$	0.019 h/Mpc	0.02	6.60%
Correlation r_0	$A_5 \cdot F_{\text{TRZ}} \cdot [\text{SSq}] \cdot (1 + K_{\text{MEX}} \cdot F_{\text{TRZ}}) / (1 - F_{\text{TRZ}})$	4.5 Mpc	5	8.17%
Spectral index n_s	$1 -$ $F_{\text{TRZ}} \cdot (1 - F_{\text{TRZ}} \cdot [\text{SSq}])$	0.906	0.965	6.15%

Complete structure formation from CMB to galaxy survey scales:

PAPER_1856 (CMB): 5 acoustic peaks + damping **PAPER_1862** (DM halos): NFW c , subhalos, satellites **PAPER_1867** (CvB): $T = 1.945$ K, N_{eff} **PAPER_1871 (this): $P(k)$, σ_8 , $\xi(r)$, halo bias, spectral index**

Cross-scale cosmological framework complete.

Summary Table

Complete Structure Formation Sector

Observable	UQFF	Data	Residual
σ_8	0.808	0.811	0.37% $\square\square$
BAO scale (Mpc)	145.2	147	1.22% $\square$
Correlation γ	1.843	1.8	2.39% $\square$
Halo α	1.943	1.9	2.26%
k_{peak} (h/Mpc)	0.0187	0.02	6.60%
Correlation r_0	4.59	5.0	8.17%
Spectral n_s	0.906	0.965	6.15%
Growth factor f	0.502	0.53	5.2%

Comparison Across Frameworks

Framework	Free params	Verdict
UQFF (this paper)	0	0.37-8.17% multi-observable
Λ CDM (Planck)	6	0.1% fits via parameter freedom
Non-standard cosmology	many	model-dependent
Modified gravity	3-5	fits some

UQFF Derivation

$\sigma_8 = 0.808 \square\square$ (from PAPER_1829)

Extends CC2 cosmology. $\sigma_8 = 0.808$ vs Planck 2018 = 0.811 $\rightarrow$ **0.37% match**

BAO Scale $\square$

From PAPER_1856 CMB peaks:

$$r_{s_BAO_UQFF} = 145.2 \text{ Mpc}$$

vs 147 Mpc $\rightarrow$ **1.22% match**

Deep unity: same physics (sound horizon at recombination) sets CMB peaks and BAO galaxy correlation.

Galaxy Correlation Slope $\gamma \square$

$$\begin{aligned} \gamma_{UQFF} &= 2 - F_{TRZ} \cdot (K_{MEX} + K_{MEX} \cdot [SSq]) / K_{MEX} \\ &= 2 - 0.1 \cdot 3.271/2.083 \\ &= 1.843 \end{aligned}$$

vs observed $\gamma = 1.8 \rightarrow$ **2.39% match**

Physical meaning: correlation function $\xi(r) \propto r^{(-\gamma)}$. $\gamma = 1.843$ IS 2 minus F_{TRZ} vacuum-decoherence correction.

Matter Power Spectrum Peak k_{peak}

$$\begin{aligned}k_{\text{peak_UQFF}} &= F_{\text{TRZ}} \cdot (K_{\text{MEX}} - 1) \cdot [\text{SSq}] \cdot (1+F_{\text{TRZ}})^2 / D_{\text{phys}} \\&= 0.1 \cdot 1.083 \cdot 0.6897 / 4 \\&= 0.0187 \text{ h/Mpc}\end{aligned}$$

vs 0.02 h/Mpc → **6.60% match**

Peak location set by horizon at matter-radiation equality — UQFF-derived from primitives.

Halo Mass Function α (from PAPER_1862)

$$\alpha_{\text{halo_UQFF}} = 2 - F_{\text{TRZ}} \cdot [\text{SSq}] = 1.943$$

vs observed ~ 1.9 → **2.26% match** (from PAPER_1862 discovery)

Galaxy Correlation Length r_0

$$\begin{aligned}r_{0_UQFF} &= A_5 \cdot F_{\text{TRZ}} \cdot [\text{SSq}] \cdot (1+K_{\text{MEX}} \cdot F_{\text{TRZ}}) / (1-F_{\text{TRZ}}) \\&= 60 \cdot 0.1 \cdot 0.57 \cdot 1.208 / 0.9 \\&= 4.59 \text{ h}^{-1} \text{ Mpc}\end{aligned}$$

vs observed $5 \text{ h}^{-1} \text{ Mpc}$ → 8.17%.

Spectral Index n_s

$$\begin{aligned}n_{s_UQFF} &= 1 - F_{\text{TRZ}} \cdot (1 - F_{\text{TRZ}} \cdot [\text{SSq}]) \\&= 1 - 0.1 \cdot 0.943 \\&= 0.906\end{aligned}$$

vs Planck 0.965 → 6.15%.

Physical Mechanism: F_{UBi} Structure Formation

Standard Λ CDM: primordial perturbations + gravitational instability + dark matter drag → structure.

UQFF picture: 1. **Primordial perturbations** from inflation (PAPER_1825, r-parameter) 2. **F_{UBi} buoyancy** provides structure formation without dark matter 3. **Same F_{UBi} from galactic rotation** (PAPER_1855), solar system (PAPER_1860), DM halos (PAPER_1862) 4. **Universal mechanism** — no need for dark matter particle

Complete cosmology sector across UQFF: - Cosmological Λ (PAPER_1156) — 0.003% - CMB peaks (PAPER_1856) — 0.31-1.83% - DM halos (PAPER_1862) — subhalo α 0% EXACT - C ν B (PAPER_1867) — N_{eff} 0.086% - σ_8 (PAPER_1829) — 0.37% □ - **Structure formation (this)** — multi-observable

Bonus Predictions

Void Statistics

Voids peak at ~ 50 Mpc radius. UQFF: $r_{\text{void}} = D_{\text{crit}} \cdot K_{\text{MEX}} \cdot (1+F_{\text{TRZ}}) / D_{\text{phys}} = 15 \text{ Mpc}$ (too small — needs refinement)

Redshift-Space Distortions

$f = \Omega_m^{0.55} \approx 0.53$ UQFF ≈ 0.50 — consistent.

Higher-Order Correlations

- $\xi(r)$ at large r
- Higher-order n -point correlations
- Bispectrum $B(k)$

Void-Halo Cross-Correlation

Standard: negative correlation UQFF: predicted consistent

DESI + Euclid Precision Tests

DESI (2024+) and Euclid (2024+) will measure $P(k)$, $\xi(r)$ to $<1\%$ precision. - Test UQFF k_{peak} at improved precision - Test UQFF σ_8 evolution with redshift

Falsifiability Statements

Immediate:

1. **DESI Y1 (2024) results** — first-year galaxy correlation.
 - Test UQFF $r_0 = 4.59$ vs 5
2. **Euclid preliminary (2024-2025)** — cluster counts, weak lensing.
 - Test UQFF halo mass function $\alpha = 1.943$

Longer-term (2028+):

3. **Euclid final + Roman (2027+)** — full survey.
 - Test UQFF σ_8 at multiple redshifts
 - Test UQFF k_{peak} at higher precision

Structural falsifiers:

- If σ_8 measured significantly $\neq 0.808$: UQFF formula wrong
- If γ measured $\neq 1.84$: $F_{\text{TRZ}} \cdot (K_{\text{MEX}} + K_{\text{MEX}} \cdot [\text{SSq}]) / K_{\text{MEX}}$ wrong

Cross-References

- **PAPER_646** — Universal Inertial Operator (foundational)
- **PAPER_1023** — Neutrino PMNS (foundational)
- **PAPER_1065** — F_{UBi} buoyancy
- **PAPER_1156** — CC2 cosmology (Λ base)
- **PAPER_1802** — $D_{\text{crit-26}}$ polynomial
- **PAPER_1810** — 26th-order F_{U} master equation
- **PAPER_1829** — **σ_8 tension resolution** □
- **PAPER_1830** — JWST early galaxies
- **PAPER_1855** — Galactic rotation (F_{UBi})
- **PAPER_1856** — **CMB acoustic peaks** □
- **PAPER_1862** — **DM halo alternative (subhalo $\alpha = 1.9$ EXACT)** □
- **PAPER_1867** — $C_{\nu B}$ (N_{eff})

NOT REPLACEMENT

Standard Λ CDM + N-body simulations + linear perturbation theory provide baseline for structure formation. UQFF adds first-principles derivation of σ_8 , BAO, correlation function, halo bias via F_{UBi} buoyancy + primitive combinations at zero free parameters. Residuals reported honestly per Rule 7.

If DESI/Euclid precision measurements reveal significant deviations, formulas require revision.

Reference

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- **DESI Collaboration** (2016). *The DESI Experiment*. arXiv:1611.00036
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- Companion UQFF whitepapers: PAPER_646, PAPER_1023, PAPER_1065, PAPER_1156, PAPER_1802, PAPER_1810, PAPER_1829, PAPER_1830, PAPER_1855, PAPER_1856, PAPER_1862, PAPER_1867

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